🗺️ Technical Roadmap: Pro Quant Terminal (v2.0)

From Data Engineering to Wealth Management Product

🎯 Vision Statement

Transition from a "script-based" ledger parser to a **State-Persistent Multi-Portfolio System**. The tool will evolve from a retrospective reporter into an active strategic advisor, allowing for "What-If" simulations, multi-account management (ISA/SIPP), and automated AI risk monitoring.

🏗️ Phase 1: The Multi-Account Foundation (Persistence)

Goal: Stop re-uploading files. Create a permanent home for your ISA and SIPP.

- **Database Migration:** Implement a Google Sheets backend.
- **The Persistence Layer:** Ledgers are uploaded once and stored. The tool "remembers" your holdings, historical cleanup, and manual ticker overrides.
- **Account Switching:** A top-level toggle to view **[ISA]**, **[SIPP]**, or **[Consolidated Master View]**.
- **Tax Relief Logic:** Specific to the SIPP, add logic to track "Gross vs. Net" investment to account for the government tax top-ups.

🧪 Phase 2: The "Shadow" Sandbox (Simulation)

Goal: Mathematically evaluate a trade before you execute it.

- **Pro-Forma Trades:** A "Simulation Mode" where you can add "Paper Trades" to your current live holdings.
- **Impact Analytics:**
    - **Beta Shift:** "How does buying this ETF change my sensitivity to the S&P 500?"
    - **Concentration Alert:** Automated check for overlap (e.g., "You are adding more Apple exposure via this new ETF").
    - **Volatility Projection:** Uses the last 3 years of correlations to see if the new asset diversifies or amplifies portfolio risk.

🤖 Phase 3: The AI Concierge & Risk Sentry

Goal: Passive monitoring and professional-grade reporting.

- **The "Investor Letter":** A Gemini-powered module that runs every Friday/Quarter. It ingests your TWR data and writes a clinical, institutional-style performance review.

- **Signal Sweep:** An automated daily check of technical levels (e.g., "Ticker X has dropped below its 200-day Moving Average").
- **Flash Alerts:** Dashboard notifications for high-volatility events or VaR (Value at Risk) breaches.

🛑 The "Stable Engine" Covenant (Warning)

The core **Ledger-Parsing Logic** and **Pence/Pound Cleanse Algorithm** are now deemed **Stable**.

> *Developer Note:* Future iterations on UI or AI must interact with the *output* of the engine. Do not modify the internal math of the Time-Weighted Return (TWR) or the Multiplier Logic without a full regression test against the Barclays Smart Investor baseline.

🖼️ Visual Architecture Overview

I've rendered the following "System Blueprint" for you to keep as a reference:

